

Model Outputs, Results, and Interpretation Summary

Problem Statement Alignment

This investigation addresses the critical question:

"Which wards in KwaZulu-Natal are at greatest risk of catastrophic low voter turnout in the 2026 Local Government Elections, and how can resources be targeted to reverse civic disengagement?"

In the 2021 Local Government Elections, KZN turnout collapsed from **61.4% (2016)** to **49.5% (2021)**, leaving millions of registered electors disengaged.

Key Model Output Highlights (2026 Forecasts Across 921 Wards)

- 1. Total Wards Forecasted:** 921 wards across all 44 municipalities in KwaZulu-Natal.
- 2. Mean Projected Turnout (2026): 60.90%** (an expected recovery of approximately 11.4 percentage points relative to the depressed 2021 baseline).
- 3. Total Estimated Ballots to be Cast: 3,636,162 votes** from an official registered roll of **6,030,969 voters**.
- 4. Participation Risk Stratification:**
 - **Substantial Increase (>5pp):** 869 Wards (Communities with strong historical potential for re-engagement)
 - **Moderate Increase (1–5pp):** 48 Wards (Gradual stabilization)
 - **Stable (±1pp):** 2 Wards
 - **Moderate Decline (1–5pp):** 2 Wards (High-priority civic intervention targets)

Output File Inventory

- **kzn_ward_turnout_predictions_2026.csv:** Full 921-ward prediction table with registered voters, predicted turnout %, expected vote volume, and risk category.
- **model_evaluation_benchmark_scorecard.csv:** Comparative evaluation of Baseline (MAE 11.14), Winning Model (MAE 10.14, +8.99% gain), and Rejected Model (MAE 11.48).
- **feature_importance_ranking.csv:** Feature weights proving past voting habit drives >70% of prediction accuracy.